

CARDIFF, United Kingdom

WMO: 037150

Lat: 51.397N

Lon: 3.343W

Elev: 220

StdP: 14.58

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	28.2	30.2	21.3	16.1	31.4	24.4	18.7	33.2	31.2	45.5	28.8	45.2	10.2	70	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	75.6	64.1	72.0	62.4	69.7	61.6	65.6	72.3	64.2	69.6	63.1	67.8	9.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	86.9	67.1	62.5	85.2	66.5	61.0	80.8	65.0	30.5	72.8	29.4	69.3	28.6	67.4	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	23.7	21.1	24.1	81.5	2.8	3.8	22.0	84.2	20.4	86.4	18.8	88.6	16.8	91.3
			22.9	68.7	2.8	1.9	20.9	70.1	19.3	71.2	17.7	72.3	15.7	73.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	41.9	42.0	44.5	48.2	53.4	58.4	62.0	61.7	58.3	53.3	47.2	43.0	(6)
(7)		DBStd	8.71	5.40	4.90	4.58	4.64	4.62	4.13	4.13	3.71	3.65	4.69	5.28	6.10	(7)
(8)		HDD50	1114	251	224	177	86	17	0	0	0	0	23	113	223	(8)
(9)		HDD65	5078	715	643	637	505	361	204	114	117	204	362	535	681	(9)
(10)		CDD50	1555	1	1	5	30	122	252	371	363	249	126	28	7	(10)
(11)		CDD65	43	0	0	0	0	1	6	20	14	2	0	0	0	(11)
(12)		CDH74	202	0	0	0	0	4	40	105	51	2	0	0	0	(12)
(13)		CDH80	27	0	0	0	0	0	5	14	8	0	0	0	0	(13)
(14)	Wind	WSAvg	10.8	12.2	11.6	11.4	10.3	10.6	10.0	9.9	9.7	10.1	11.1	10.8	11.8	(14)
(15)	Precipitation	PrecAvg	37.3	3.6	2.5	2.8	2.2	2.4	2.4	2.8	3.2	3.5	3.8	4.1	3.9	(15)
(16)		PrecMax	50.2	6.8	5.4	6.3	4.8	6.1	5.9	6.6	7.9	8.8	8.4	8.5	8.2	(16)
(17)		PrecMin	27.2	0.4	0.0	0.5	0.4	0.2	0.2	0.5	0.9	1.0	0.4	1.5	1.2	(17)
(18)		PrecStd	5.0	1.7	1.4	1.4	1.1	1.4	1.4	1.5	1.7	1.9	1.9	1.9	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	53.9	59.3	68.2	73.5	79.1	82.1	80.2	73.2	65.8	59.0	55.8	(19)
(20)		MCWB	51.5	50.3	51.3	56.8	61.0	63.8	65.4	66.1	64.1	61.3	56.6	53.5	(20)	
(21)		2%	DB	52.0	51.8	55.2	62.3	68.3	73.2	76.9	73.6	68.4	62.7	57.3	54.0	(21)
(22)		MCWB	50.0	49.0	49.8	53.7	59.4	62.2	64.3	64.4	61.5	59.0	55.2	52.1	(22)	
(23)		5%	DB	50.3	50.1	52.2	57.6	64.6	69.1	73.1	70.1	66.3	61.0	55.7	53.2	(23)
(24)		MCWB	49.0	48.1	48.3	51.4	57.3	60.6	62.8	62.6	60.6	58.0	53.6	51.4	(24)	
(25)		10%	DB	49.6	48.6	50.4	55.3	61.0	66.0	69.7	68.0	64.5	59.4	54.6	51.6	(25)
(26)		MCWB	48.4	47.1	47.8	50.2	55.3	59.2	61.8	61.5	59.6	56.6	52.5	49.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.4	51.3	52.9	57.5	63.5	67.1	67.5	67.9	65.4	61.9	57.5	54.5	(27)
(28)		MADB	53.5	53.5	57.3	64.9	69.7	75.8	77.9	75.1	70.1	64.5	58.5	55.8	(28)	
(29)		2%	WB	50.8	50.0	51.1	54.7	60.2	63.4	65.5	65.6	63.5	60.1	55.8	53.1	(29)
(30)		MADB	51.7	51.0	54.0	60.6	66.3	69.9	73.5	70.9	67.2	62.0	57.1	54.4	(30)	
(31)		5%	WB	49.4	48.5	49.9	52.7	58.3	61.5	64.0	64.0	61.9	59.0	54.5	51.8	(31)
(32)		MADB	50.2	49.4	51.4	57.2	63.5	67.1	70.2	68.2	65.2	60.8	55.9	53.0	(32)	
(33)		10%	WB	48.1	47.4	48.6	51.1	56.3	59.9	62.7	63.0	60.5	57.6	52.9	50.2	(33)
(34)		MADB	48.9	48.4	50.1	54.4	60.5	64.7	67.7	66.9	63.3	59.6	54.6	51.2	(34)	
(35)	Mean Daily Temperature Range	MDBR	7.8	8.7	10.1	12.0	12.3	12.3	12.5	11.9	11.5	9.4	8.5	7.8	(35)	
(36)		5% DB	MCDBR	7.8	8.7	12.0	15.7	17.5	18.4	18.7	15.2	13.5	9.3	7.7	7.5	(36)
(37)		MCWBR	7.8	7.2	8.5	9.8	10.2	10.1	8.9	8.1	8.5	7.3	7.0	7.5	(37)	
(38)		5% WB	MCDBR	7.6	8.0	9.9	13.7	15.2	15.1	15.1	12.7	11.1	8.3	7.1	7.1	(38)
(39)		MCWBR	8.0	7.5	8.1	9.3	9.6	9.1	8.5	7.9	8.2	7.5	7.3	7.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.329	0.343	0.374	0.392	0.397	0.388	0.387	0.373	0.365	0.353	0.328	0.319	(40)	
(41)		taud	2.355	2.383	2.302	2.274	2.298	2.356	2.363	2.418	2.433	2.437	2.409	2.330	(41)	
(42)		Ebn at Noon	214	246	259	268	272	275	273	272	260	240	214	196	(42)	
(43)		Edh at Noon	20	26	33	38	39	37	36	33	29	24	19	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	249	470	834	1286	1578	1713	1605	1331	1003	567	301	194	(44)	
(45)		RadStd	23	57	106	144	80	180	148	116	73	49	29	24	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page